Bill of materials

Resistors	
Part	**Value**
R-LED	4k7
R1	1m
R2	1m
R3	10k
R4	100k
R5	1m
R6	3k3
R7	200k
R8	470k
R9	1m
R10	1m
R11	470k
R12	6k8
R13	1m
R14	22k
R15	330k
R16	6k8
R17	3k3
R18	330k
R19	1k
R20	10k
R21	1m
R22	100k
R23	33k
R24	10k
R25	22k
R26	22k
R27	47k
R28	10k
R29	22k
R30	10k
R31	10k
R32	1k
R33	10k
R34	10k
R35	10k
R36	3k3
R37	1m
R38	2k2
R39	2k2
R40	220k
R41	220k
R42	2k2
R43	2k2
R44	220k
R45	220k
R46	100k
R47	1k
R48	1k
R49	10k
R50	10k
R51	1k
R52	100k
R53	100k
R54	10k

Capacitors	
Part	**Value**
C1	100n
C2	1n
C3	220n
C4	22n
C5	22n
C6	22n
C7	100n
C8	220pf
C9	22n
C10	47pf
C11	4n7
C12	47n
C13	220n

C14	220pf
C16	680pf
C17	22n
C18	1n
C19	2n2
C20	1n
C21	100n
C23	330n
C25	22n
C26	22n
C27	1n
C28	10n
C29	10n
C30	47p
C32	22n
C33	22n
C34	6n8
C35	680pF
C38	1uF, 50V
C39	1uF, 50V

Electrolytic Capacitors	
Part	**Value**
C15	2u2
C22	100uf
C24	100u
C31	2u2
C36	2u2
C37	2u2

Potentiometers	
Part	**Value**
BLEND	100K B
DRIVE	100K C
HI	100K B
HI-MIDS	100K B
LEVEL	100K A
LO	100K B
LO-MID	100K B

IC	
Part	**Value**
IC1	TL072ACP
IC2	TL072ACP
IC3	4049N
IC4	TL072ACP
IC5	TL074ACN
IC6	TL074ACN

Transistors	
Part	**Value**
Q1	J201
Q2	J201

Switches	
Part	**Value**
Ultra Hi	SPDT On-On
Grunt	SPDT On-Off-On
Ground-lift	SPDT On-On
-	3PDT Stomp foot

Jacks	
Part	**Value**
X2R JACK *	X2R
-	DC JACK
-	AUDIO JACK
-	AUDIO JACK

Diodes	
Part	**Value**
D1	1n4148
D2	1n4148
D3	1n5817
D4	3mm red LED

Ultra-Mod

After several tests with our Black Mirror Seven, we decided to try the ultra-mods and add two toggles that switch between the stock capacitor (C33, C35) and some suggested values to experiment with different frequencies responses.
Check the tables below to have a reference on which capacitor to use to boost the desired frequencies.
Take in mind that this type of Baxandal EQ always has a pretty low Q factor, in this case around 0.4. So, for low amounts of boost or cut, it acts more like a volume control.
It only highlights the mentioned frequencies at full boost or full cut. It acts as a funny volume control in all other positions since it is too wide band and not selective.

These toggles might be hard to fit in such a tight project as this one. We will realize a new project that includes these boards in late 2021.
For doing this mod, you must **remove C33 and C35** from the PCB and place them as indicated in the diagram below.
Choose the capacitors of your preference for CX and CY following the table above.
Capacitors are soldered on the switch; take special care not to overheat the pads and melt the switch.

Thanks to Vivek Mehta for helping us out with these mods!

Low mids Switch			
Range in Ultra: 1khz - 500hz			
Part	**Value**	**Freq**	**Decibels**
CX	100n F	168 Hz	26 dB
CX	47n F	236 Nz	25 dB
CX	22n F	340 Hz	24 dB
C33	**10n F**	**500 Hz**	**21 dB**
CX	5n F	708 Hz	18 dB

Hi mids Switch			
Range in Ultra: 3khz - 1.5khz			
Part	**Value**	**Freq**	**Decibels**
CY	3300pf F	1.55 Khz G	23 dB
CY	2200pf F	1.9 Khz G	21 dB
CY	1000pf F	2.8 Khz G	16.6 dB
C35	**680pf F**	**3.3 Khz G**	**14 dB**
CY	500pf F	4 Khz G	12.6 dB

Schematic